

Shortest Subsequence

Time limit: 1.00 s

Memory limit: 512 MB

You are given a DNA sequence consisting of characters A, C, G, and T.

Your task is to find the shortest DNA sequence that is not a subsequence of the original sequence.

Input

The only input line contains a DNA sequence with n characters.

Output

Print the shortest DNA sequence that is not a subsequence of the original sequence. If there are several solutions, you may print any of them.

Constraints

- $1 \leq n \leq 10^6$

Example

Input	Output
ACGTACGT	AAA